

AUTOMATED RECIPE GENERATOR BASED ON LEFTOVERS

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ABSTRACT:

In the era of increasing food waste and fast-paced lifestyles, many households discard edible leftovers due to a lack of ideas on how to reuse them effectively. To address this issue, we propose a Smart Leftover-Based Recipe Generator, a full-stack web application designed to assist users in preparing dishes using available ingredients without the use of machine learning. This system allows users to select a category—Vegetarian or Non-Vegetarian—and choose multiple leftover ingredients from a predefined list. Based on the selected ingredients, the application dynamically fetches and displays relevant recipes with proper preparation steps, including the addition of spices and other

essentials, by integrating third-party recipe APIs. The frontend of the application is developed using React.js and Tailwind CSS, offering a responsive and user-friendly interface. The backend, built with Node.js and Express.js, handles API requests, manages user sessions, and facilitates communication with the database. MongoDB is used for storing user credentials and maintaining the history of generated recipes, allowing users to revisit past suggestions. The authentication is handled via JWT (JSON Web Token) to ensure secure access. One of the key features of the system is that it does not store actual recipes in the database, minimizing storage requirements. Instead, it stores user selections and history references,

thus, enabling efficient tracking without redundancy. The application is deployed using modern cloud platforms—Vercel for frontend, Render for backend, and MongoDB Atlas for the database—ensuring scalability and high availability. This solution promotes sustainability by minimizing food waste, simplifies daily cooking decisions, and provides an easy-to-use platform even for users with minimal culinary knowledge. The project proves that practical, real-world applications can be developed using lightweight full-stack technologies without relying on complex machine learning models.

INTRODUCTION:

Food waste is a pressing global issue, with millions of tons of edible items being discarded daily due to improper planning or lack of culinary ideas. A significant portion of this waste comes from leftover ingredients that are often ignored or thrown away. At the same time, many individuals struggle to decide what to cook with limited ingredients, especially when time or cooking knowledge is limited. This challenge highlights the need for a practical, user-friendly solution that can convert available leftovers into complete, enjoyable meals. To address this, we present a Smart Leftover-Based Recipe Generator, a full stack web application that assists users in generating personalized recipes based on the ingredients they have on hand. Unlike traditional cooking apps that rely on static recipe lists or complex machine learning models, our system uses API-based dynamic recipe generation to provide accurate and relevant suggestions instantly. The application is designed to be accessible and efficient, requiring no prior login to access its core features, while also

supporting a secure login system for users who wish to save their history. The system is built using modern technologies—React.js and Tailwind CSS for the frontend, Node.js and Express.js for the backend, and MongoDB for data persistence. This project demonstrates how a lightweight yet powerful tech stack can be utilized to solve everyday problems in a sustainable way. By turning leftover ingredients into practical meals, the application encourages responsible food usage, reduces waste, and simplifies the cooking experience for all users.

RELATED WORKS:

Several existing applications and platforms focus on recipe generation and food waste reduction. Applications such as SuperCook, BigOven, and Cookpad allow users to input ingredients and receive recipe suggestions. However, many of these rely on static databases or require extensive user input, and some limit access to premium features, reducing usability for free users. Moreover, these platforms often lack dynamic generation based on leftover-focused use cases and do not always provide step-by-step cooking instructions tailored to selected ingredients.

Some academic works have explored using machine learning and recommendation systems for recipe suggestions, focusing on user preferences, ingredient popularity, or nutrition optimization. While effective, these approaches often require large datasets and complex training models, which increase system complexity and reduce accessibility for simple use cases.

In contrast, our system adopts a lightweight yet dynamic approach by

integrating third-party APIs for recipe generation. It avoids ML complexity while maintaining variety and accuracy in recipe suggestions. Additionally, by incorporating user login and history tracking, our application bridges the gap between usability and personalization, offering a more focused solution for real-world leftover utilization. This makes it both scalable and suitable for everyday users who seek quick, practical, and sustainable cooking ideas.

PROPOSED METHOD:

The proposed system is a full-stack web application designed to generate recipes based on leftover ingredients provided by users. It follows a modular structure that ensures simplicity, scalability, and user convenience without involving complex machine learning models. The major components of the system are as follows:

1. Ingredient Selection Interface

Users begin by selecting the type of recipe—Vegetarian or Non-Vegetarian. Based on this choice, a dynamic dropdown menu allows users to select one or more leftover ingredients. The interface is built using React.js and styled with Tailwind CSS, ensuring a smooth and responsive user experience.

2. Recipe Generation via API

Once the ingredients are selected, the application sends a request to a third-party recipe API. This API returns recipes that match the chosen ingredients. The system parses the response and displays a complete recipe, including ingredients, preparation steps, and spice additions, tailored to the selection.

3. User Authentication and Session Management

To allow users to track their activity and access previous recipe suggestions, a secure login and logout system is implemented using JWT (JSON Web Token). Users can register, sign in, and have their activity securely recorded and associated with their account.

4. Recipe History Tracking

Every recipe generated by a logged-in user is recorded in the database, but only the selection data (ingredients and time) is stored—not the full recipe—ensuring lightweight storage. The user can revisit this history to regenerate or review past suggestions.

5. Backend Logic and API Integration

The backend of the recipe generator web application is developed using Node.js with the Express.js framework to ensure fast and scalable handling of HTTP requests. It is responsible for managing all routing and server-side logic. When users interact with the frontend—for example, by selecting ingredients—the backend routes these requests to appropriate endpoints. JWT (JSON Web Token) is used to manage user authentication and ensure secure access to protected routes. Additionally, the backend integrates with a third-party recipe API to fetch relevant and real-time recipe suggestions based on the selected leftover ingredients. This approach ensures that the application offers diverse and up-to-date cooking suggestions while keeping the backend logic light and responsive. Data related to users and their past interactions is securely stored in the connected MongoDB.

database for personalized user experiences.

6. Database Management

The project uses MongoDB as the primary database for efficient, scalable, and document-based data storage. MongoDB is employed to store essential user-related information, including login credentials and session tokens, enabling secure and persistent authentication. It also saves the selected ingredients for each user to maintain a history of generated recipes, which can be retrieved and displayed to provide a personalized experience. Furthermore, session-related data is stored to handle currently logged-in users and manage their activities. The database schema is designed to be flexible yet optimized, ensuring that all read and write operations are performed with minimal latency, even when the data grows with increased user activity.

7. Deployment Strategy

The deployment architecture of the application is designed to be lightweight, secure, and highly accessible. The frontend, built using React.js and styled with Tailwind CSS, is deployed on Vercel, a platform known for its seamless integration with Git and high-performance static site hosting. This enables quick rendering of the user interface with minimal load times. The backend services, powered by Node.js and Express, are deployed on Render, which offers robust handling of dynamic API routes and background server operations. MongoDB Atlas is used as the cloud-hosted database solution, ensuring secure and scalable data management. With this deployment strategy, the entire application remains

available, fast, and secure for all users across devices and locations.

This method ensures the application remains lightweight, scalable, easy to maintain, and user-friendly, while effectively addressing the problem of food wastage through intelligent recipe generation using available leftovers.

EXPERIMENTAL RESULTS AND DISCUSSION:

1. User Testing and Functionality Evaluation

The proposed system was tested across multiple devices and browsers by different users to ensure its compatibility and usability. Users were asked to select ingredients from the dropdown list under either the vegetarian or non-vegetarian category. Upon submission, the system successfully fetched relevant recipes through the integrated API, displaying complete instructions including ingredients, spice additions, and cooking steps. The frontend interface was responsive and user-friendly, providing smooth navigation even for first-time users with minimal technical knowledge.

2. Recipe Accuracy and Relevance

The generated recipes were evaluated for culinary relevance and variety. For combinations like *rice + vegetables*, the system returned dishes such as *vegetable pulao* and *fried rice*. For *chicken + onion*, it generated recipes like *chicken curry* and *pepper chicken*. These outputs were found to be accurate, contextually appropriate, and practical for everyday cooking. Since the system relies on real-time API

responses, the suggestions remained dynamic and diverse across repeated tests, enhancing user engagement.

3. Performance and Responsiveness

The system maintained excellent performance throughout testing. On average, recipe generation took less than three seconds, even during concurrent requests. The backend, built with Node.js and Express.js, handled routing and API calls efficiently without significant load time. MongoDB operations, including storing and retrieving history data, were also executed quickly and reliably. The JWT-based authentication ensured secure login and session handling without affecting system performance.

4. User History and Data Retrieval

The history feature worked as intended, allowing users to revisit their previously selected combinations and view the corresponding recipes again without re-entering data. As only selection data and timestamps were stored, the database remained lightweight and efficient. This functionality provided personalization without storing the actual recipe content, which aligns with the application's goal of minimizing storage while maximizing utility.

5. Summary of Results

The experimental results confirm that the application performs effectively in real-world usage scenarios. It successfully addresses the problem of leftover ingredient utilization by offering reliable, personalized, and efficient cooking solutions. The system's lightweight

architecture, dynamic content generation, and seamless user experience demonstrate its potential for everyday use and future scalability.

CONCLUSION:

The Automated Recipe Generator Based on Leftovers presents an innovative yet straightforward solution to a common problem faced by households—how to make use of available ingredients efficiently. By allowing users to select leftover items and generating appropriate recipes based on their choices, the system minimizes food waste and encourages sustainable cooking habits. The project combines a user-friendly frontend built with React.js and Tailwind CSS with a robust backend using Node.js, Express.js, and MongoDB for secure user management and data storage. Integrating a third-party recipe API eliminates the need for a complex dataset or machine learning model, while still delivering a wide range of reliable and diverse recipes. The system's deployment on Vercel, Render, and MongoDB Atlas ensures a fast, scalable, and accessible web application. Ultimately, this project not only enhances the cooking experience but also aligns with broader goals of sustainability, digital convenience, and intelligent food utilization.

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